hJLP: Delta-Hedged Jupiter Liquidity Provider Strategy

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Abstract

This whitepaper introduces hJLP (Hedged Jupiter Liquidity Provider), a sophisticated risk management strategy developed by Gauntlet for the Jupiter perpetual exchange on the Solana blockchain. hJLP aims to mitigate the inherent price risks faced by liquidity providers while maintaining attractive returns. By implementing hedging strategies, hJLP seeks to reduce the volatility of returns for liquidity providers, thereby improving the risk-adjusted performance of the strategy.

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1 Introduction

1.1 Background

The rapid evolution of decentralized finance (DeFi) continues to present both opportunities and challenges for market participants. As the industry matures, there's a growing need to bridge the gap between the innovative potential of blockchain technology and the robust risk management practices of traditional finance.

1.2 The Perpetual Exchange Landscape

Perpetual exchanges have become a cornerstone of the cryptocurrency trading ecosystem. Unlike spot exchanges or traditional Automated Market Makers (AMMs), perpetual exchanges offer derivative contracts that don't have an expiry date, allowing traders to hold positions indefinitely. These exchanges have gained popularity due to their ability to offer high leverage and the ease of going both long and short on assets.

1.3 Jupiter: A Solana-based Perpetual Exchange

Jupiter stands out as a prominent perpetual exchange built on the Solana blockchain. Leveraging Solana's high throughput and low transaction costs, Jupiter offers a robust trading experience with deep liquidity. The platform utilizes a unique liquidity pool design, where liquidity providers can contribute assets and earn fees from trading activity. Jupiter's protocol demonstrates key characteristics of successful decentralized finance (DeFi) platforms:

1. Market Dominance: With AUM exceeding $600M, Jupiter has achieved significant market share, potentially benefiting from first-mover advantages or superior product-market fit in the perpetual futures niche.

2. Liquidity Depth: The 30-day average volume data shows sustained high liquidity, crucial for efficient price discovery and reduced slippage in perpetual futures markets.

3. JLP Yield: The historical JLP APY data reveals convergence towards a 30-60% range, suggesting an equilibrium between capital attraction and sustainable returns. This stability is vital for long-term viability in DeFi protocols.

1.4 The Role of Liquidity Providers

In the context of Jupiter's perpetual exchange, liquidity providers play a crucial role. They contribute assets to the Jupiter Liquidity Provider (JLP) Pool, which acts as a counterparty to traders. When traders seek to open leveraged positions, they borrow tokens from this pool.

The JLP token represents a share in this liquidity pool, and its value is derived from:

1. An index fund of SOL, ETH, WBTC, USDC, and USDT
2. Traders' profit and loss
3. 75% of the generated fees from opening and closing positions, price impact, borrowing fees, and trading fees

1.5 The Need for Enhanced Risk Management

Despite the potential for earning trading fees, the risks associated with being a liquidity provider can be substantial, especially in the volatile cryptocurrency market. Traditional risk management techniques from the world of finance could potentially mitigate these risks, but they need to be adapted to the unique characteristics of perpetual crypto exchanges.

This is where hJLP comes in. By implementing sophisticated hedging strategies, hJLP aims to offer liquidity providers a way to earn fees from Jupiter's trading activity while minimizing exposure to price fluctuations in the underlying assets.

2 About Gauntlet

Gauntlet is a leading financial modeling and simulation platform for blockchain protocols and decentralized finance (DeFi) applications. Founded in 2018, Gauntlet has established itself as a pioneer in bringing sophisticated risk management techniques to the crypto ecosystem.

2.1 Company Overview

Gauntlet currently models over $35 billion for customers on-chain and has been providing risk management and optimization services for approximately six years. The company serves some of the largest DAOs, projects, and foundations in the crypto space, with its work spanning various sectors including

decentralized exchanges (DEX), lending protocols, restaking mechanisms, perpetual exchanges, and entire blockchain ecosystems.

2.2 Expertise and Services

Gauntlet's core competencies include:

1. Risk modeling and management for DeFi protocols
2. Optimization of protocol parameters for enhanced performance and security
3. Stress testing and scenario analysis
4. Economic security analysis
5. Liquidity mining program design

The company has worked with prominent projects in the space, including Eigenlayer, Uniswap, Arbitrum, and Jupiter.

2.3 Gauntlet's Role in Jupiter's Ecosystem

Gauntlet has been working closely with the Jupiter team to optimize their perpetual exchange since early 2024. Some of the key contributions include:

1. Completing and publishing a comprehensive market risk methodology for Jupiter
2. Building custom Parameter Optimization and Risk Models for Jupiter
3. Continuously refitting, updating assumptions, and analyzing model outputs
4. Optimizing key metrics while controlling for open interest imbalance risk, including:
    - Maximizing protocol trading volume
    - Maximizing capital efficiency for traders
    - Maximizing JLP pool liquidity
    - Minimizing persistent skew and high utilization risk
5. Publishing a series of analyses and recommendations to optimize the above metrics
6. Providing a Risk Dashboard for real-time monitoring and analysis

Gauntlet's extensive experience in risk management and protocol optimization positions the company uniquely to develop and implement the hJLP strategy, bringing institutional-grade risk management to Jupiter's liquidity providers.

3 Theoretical Foundations: Bridging TradFi and DeFi

3.1 The Evolution of Hedging Theory

To understand hJLP, we must first take a journey through the annals of financial theory. The concept of hedging dates back to ancient times, but its modern incarnation began with the seminal work of Louis Bachelier in 1900. Bachelier's thesis, "The Theory of Speculation," laid the groundwork for much of modern financial mathematics.

Key milestones in the development of hedging theory include:

1. 1900: Louis Bachelier's "The Theory of Speculation"
2. 1952: Harry Markowitz's Modern Portfolio Theory
3. 1973: Black-Scholes-Merton options pricing model
4. 1979: Cox-Ross-Rubinstein binomial options pricing model
5. 1992: Heath-Jarrow-Morton interest rate models

The Black-Scholes-Merton model, in particular, introduced the concept of "delta," a measure of the rate of change in the option's value with respect to changes in the underlying asset's price. Delta hedging, the technique of offsetting delta risk by taking opposing positions, became a cornerstone of risk management in traditional finance.

3.2 Adapting Delta Hedging for DeFi

While the principles of delta hedging are well-established in traditional finance, applying them to the 24/7, highly volatile world of cryptocurrency markets presents unique challenges:

1. Continuous Market Operations: Unlike traditional markets with defined trading hours, crypto markets operate non-stop, requiring constant monitoring and potential rebalancing.
2. High Volatility: Cryptocurrency markets are known for their extreme price swings, which can lead to rapid changes in delta exposure.
3. Funding Rates: Perpetual futures contracts use funding rates to tether their prices to the spot market, adding an additional layer of complexity to hedging strategies.
4. Liquidity Constraints: While improving, the liquidity in crypto derivatives markets can still be limited compared to traditional markets, potentially leading to higher slippage during hedging operations.

5. Smart Contract Risks: The use of smart contracts introduces additional operational risks not present in traditional finance.

To address these challenges, the hJLP strategy incorporates several innovations:

1. Real-time Delta Calculation: A sophisticated data pipeline continuously monitors the JLP pool to compute delta per JLP token based on:
   - SOL/ETH/BTC JLP Spot Liquidity
   - JLP Long Perp Positions
   - JLP Short Perp Positions
   - JLP Supply (Mints & Redeems)
2. Multi-Asset Hedging: Given the composition of the JLP pool, the strategy needs to manage delta exposure across multiple assets simultaneously.
3. Funding Rate Optimization: The hedging strategy takes into account the impact of funding rates on the overall profitability of the positions.

4 The hJLP Strategy: Detailed Mechanics

The hJLP (Hedged Jupiter Liquidity Provider) strategy is designed to mitigate the risks associated with providing liquidity to Jupiter's perpetual exchange while maintaining attractive returns. This section delves into the intricate details of the strategy's implementation, covering data ingestion, delta calculation, hedging approaches, and execution management.

4.1 Hedging Venue: Drift Protocol

The hJLP strategy will utilize Drift Protocol as its primary hedging venue. Drift is a decentralized exchange built on the Solana blockchain, offering perpetual futures trading with key features that align well with our hedging requirements:

- **High Liquidity**: Drift consistently maintains deep liquidity pools, crucial for executing large hedging transactions with minimal slippage.
- **Cross-Margining**: Drift's cross-margining feature allows for efficient capital utilization across multiple trading pairs. This allows the strategy to deposit JLP in the pool and create perp positions.
- **On-Chain Orderbook**: The on-chain orderbook provides transparency and reduces counterparty risk.

4.2 Strategy Flow

The hJLP strategy leverages Circuit's vault infrastructure and Drift's SDK to create a seamless and efficient hedging process. The operational flow is as follows:

1. **User Deposit**: Users deposit any supported assets into the hJLP vault via the partner's website. This provides flexibility for users to contribute with various cryptocurrencies.

2. **Automatic Asset Conversion**: The deposited assets are automatically swapped to JLP (Jupiter Liquidity Provider tokens) via Jupiter's DEX. This step ensures that all user deposits are converted into the required format for the hedging strategy.

3. **Collateral Deposit**: The newly acquired JLP tokens are then deposited into Drift as collateral. This feature is planned for near-future implementation and will enable efficient use of JLP within the Drift ecosystem.

4. **Initial Hedge Positioning**: SOL, ETH, and BTC perpetual futures short positions are opened according to the current delta exposure in the JLP pool. This initial hedging step aims to neutralize the portfolio's delta risk.

5. **Dynamic Rebalancing**: The system rebalances the hedges either:

   - After a predetermined number of periods (time-based rebalancing), or
   - If the hedges deviate from the target by a specified tolerance (threshold-based rebalancing).

4.2.1 Integration with Circuit and Drift

The use of Circuit's vault infrastructure provides several advantages:

- **Security**: Circuit's battle-tested vault system offers robust security for user deposits.

- **Scalability**: The vault infrastructure is designed to handle large volumes of deposits and withdrawals efficiently.

- **Flexibility**: Support for multiple asset types simplifies the user experience and broadens accessibility.

Drift's SDK integration enables:

- **Efficient Execution**: Direct interaction with Drift's trading functions for quick and precise hedge positioning.

- **Real-time Data Access**: Immediate access to market data for accurate delta calculations and rebalancing decisions.
- **Advanced Order Types**: Utilization of Drift's sophisticated order types for optimal hedge execution.

4.3 Real-time Data Ingestion

At the core of the hJLP strategy is a sophisticated real-time data pipeline that continuously monitors the Jupiter Liquidity Pool (JLP). This pipeline is crucial for maintaining an up-to-date view of the pool's composition and risk exposure.

4.3.1 Data Sources

The pipeline ingests data from several key sources:

- JLP Pool Composition: Real-time data on the amounts of SOL, ETH, BTC, USDC, and USDT in the pool.
- Perpetual Positions: Continuous updates on long and short perpetual positions for each asset.
- JLP Token Supply: Real-time tracking of JLP token mints and burns.
- Market Data: Price feeds for all relevant assets from multiple sources to ensure reliability.

4.3.2 Data Processing

The ingested data undergoes several processing steps:

1. Data Validation: Checks for data integrity and consistency.
2. Normalization: Conversion of all asset values to a common denomination (typically USDC) for easier comparison and calculation.
3. Aggregation: Combining data from multiple sources to create a comprehensive view of the JLP pool's state.

4.4 Delta Calculation Methodology

The delta calculation is at the heart of the hJLP strategy, providing a measure of the JLP token's exposure to price movements in the underlying assets.

4.4.1 Asset-specific Delta

For each asset a in the JLP pool, the delta is calculated as:

$$\Delta_a = \frac{spotLiquidity_a - longPerp_a + shortPerp_a + undistributedFees}{jlpSupply} \tag{1}$$

Where:

- $spotLiquidity_a$ is the amount of asset a held in the JLP spot liquidity
- $longPerp_a$ is the total long perpetual positions for asset a
- $shortPerp_a$ is the total short perpetual positions for asset a
- $jlpSupply$ is the total supply of JLP tokens
- $undistributedFees$ are the fees that traders have already paid, but are sitting "in escrow", waiting to be split between JLP and protocol. We hedge these fees under the assumption that the market will price them in

4.4.2 Portfolio Delta

The total portfolio delta is calculated as:

$$\Delta_{portfolio} = \sum_a \Delta_a \tag{2}$$

4.4.3 Delta Normalization

To account for the different volatilities and correlations of the assets in the JLP pool, we introduce a normalized delta:

$$\Delta_{normalized} = \sum_a w_a \cdot \Delta_a \tag{3}$$

Where w_a is a weight assigned to each asset based on its historical volatility and correlation with other assets in the pool.

4.5 Hedging Approaches

The hJLP strategy employs multiple hedging approaches, each designed to address different market conditions and risk profiles. The strategy dynamically selects and combines these approaches based on real-time market conditions and the current state of the JLP pool.

4.5.1 Temporal Delta Hedging

This approach involves rebalancing the hedge at fixed time intervals:

- 5-minute hedge: Aggressive rebalancing for tight delta control
- Hourly hedge: Balances hedging precision with transaction costs
- Daily hedge: Lower-cost approach accepting higher intraday delta exposure

The hedging frequency is dynamically adjusted based on market volatility and trading volume.

4.5.2 Delta Threshold Hedging

This approach triggers a hedge when the absolute value of the delta exceeds predefined thresholds:

- Tight bounds (e.g., ±0.5%): For more conservative risk management
- Wide bounds (e.g., ±2%): Allows for more delta fluctuation, reducing hedging costs

The thresholds are asset-specific and dynamically adjusted based on market conditions.

4.6 Execution and Slippage Management

Efficient execution of hedging trades is crucial for the success of the hJLP strategy. The strategy employs several techniques to minimize slippage and transaction costs:

4.6.1 Order Splitting

Large hedging trades are split into smaller orders to reduce market impact:

$$n_{splits} = \left\lceil \frac{order_size}{max_single_order_size} \right\rceil \tag{4}$$

4.6.2 Time-Weighted Average Price (TWAP)

For larger orders, a TWAP algorithm is used to spread the execution over time:

$$interval = \frac{total_execution_time}{n_{splits}} \tag{5}$$

4.6.3 Liquidity-Aware Execution

The strategy adjusts its execution based on available liquidity:

$$execution_size = \min(order_size, available_liquidity \cdot max_market_impact) \tag{6}$$

Where max_market_impact is a predefined threshold to limit price impact.

4.7 Continuous Monitoring and Adjustment

The hJLP strategy includes a robust monitoring system that continuously evaluates the performance of the hedging strategy and makes necessary adjustments:

- Real-time Performance Metrics: Tracking of key performance indicators such as hedge effectiveness, slippage, and transaction costs.
- Anomaly Detection: Identification of unusual market conditions or strategy behavior that may require intervention.
- Dynamic Parameter Adjustment: Automatic tuning of strategy parameters based on recent performance and market conditions.
- Circuit Breakers: Implementation of safety mechanisms to pause or adjust the strategy during extreme market events.

This comprehensive approach to strategy implementation, combining sophisticated delta calculations, multiple hedging approaches, efficient execution, and continuous monitoring, allows hJLP to provide robust risk management for Jupiter liquidity providers across a wide range of market conditions.

5 Simulation and Backtesting Methodology

To rigorously evaluate the effectiveness of the hJLP strategy, we developed a comprehensive simulation and backtesting framework. This section details our approach to modeling the complex dynamics of the Jupiter perpetual exchange and the broader cryptocurrency market.

5.1 Simulation Engine

Our proprietary simulation engine is designed to accurately replicate the behavior of the Jupiter perpetual exchange, including the JLP pool, trader activities, and market conditions.

5.1.1 Key Components

The simulation engine incorporates the following key components:

1. Asset Price Simulation: Stochastic processes modeling the price movements of SOL, ETH, BTC, and other relevant assets.
2. Order Book Simulation: Dynamic order book modeling, including liquidity depth and market impact.
3. Trader Behavior Modeling: Simulated trading activities based on historical patterns and randomized behaviors.
4. JLP Pool Dynamics: Accurate representation of JLP token minting, burning, and rebalancing.
5. Funding Rate Mechanism: Implementation of Jupiter's funding rate calculations and payments.
6. Transaction Cost Modeling: Inclusion of gas fees, slippage, and other transaction costs.

5.2 Historical Data

Our backtesting relies on a comprehensive dataset of historical market data:

- Time Period: June 1, 2024, to September 4, 2024
- Data Frequency: 1-minute intervals for price data, 5-minute intervals for order book snapshots
- Assets Covered: SOL, ETH, BTC, USDC, USDT
- Data Sources: Multiple cryptocurrency exchanges to ensure data reliability

6 Hedging Strategies

The hJLP strategy employs a range of sophisticated hedging approaches, each designed to balance risk mitigation with cost efficiency. Based on the calculated deltas and the chosen hedging strategy, the hJLP smart contract executes trades on the Drift protocol to maintain the desired hedge ratio. We have developed and tested multiple variants of the hJLP strategy to identify the most effective approach:

6.1 Unhedged JLP or Index

This serves as our baseline strategy with no hedging. It allows us to compare the performance of our hedging strategies against the unhedged position, providing a clear measure of the value added by our risk management techniques.

6.2 5-Minute Hedge

This represents our most aggressive approach to delta neutrality. Under this strategy, the hJLP smart contract rebalances every 5 minutes to fully offset the calculated delta.

Implementation:

- Calculate the current delta exposure across all assets every 5 minutes.
- Execute trades to bring the delta exposure as close to zero as possible, subject to trading costs and slippage constraints.
- Set a maximum slippage parameter (e.g., 10 bps) to prevent unfavorable executions during volatile periods.

This strategy provides the tightest control over delta exposure but may incur higher trading costs due to frequent rebalancing.

6.3 Hourly Hedge

This strategy strikes a balance between the aggressive 5-minute hedge and the more relaxed daily hedge. It rebalances on an hourly basis, potentially reducing trading costs while still maintaining relatively tight control over delta exposure.

6.4 Daily Hedge

This approach involves once-per-day rebalancing at a fixed time. It offers a lower-cost approach but accepts higher intraday delta exposure. This strategy might be suitable for less volatile assets or in markets with higher trading costs.

6.5 Bounded Delta Strategy

This strategy allows for some fluctuation in delta exposure, potentially reducing trading costs while still maintaining risk within acceptable limits. We test two variants of this approach:

6.5.1 1% Dollar Bounds

In this variant, we rebalance when delta exceeds $\pm 1\%$ of notional value. Implementation:

- Monitor delta exposure continuously.
- When delta exposure exceeds the 1% bound, execute trades to bring it back within the bound.

6.5.2 3%/4%/8% Asset-specific Bounds

This variant recognizes that different assets may have different volatility profiles and liquidity characteristics, and applies tailored hedging thresholds to each. Implementation:

- Set specific delta thresholds for each asset (e.g., 3% for BTC, 4% for ETH, 8% for SOL).
- Monitor delta exposure for each asset independently.
- When any asset's delta exposure exceeds its specific threshold, execute trades to bring it back within bounds.

This approach allows for more nuanced risk management across a diverse portfolio of assets.

6.6 Fees Only

This is a theoretical hedge that does not include transaction or slippage costs and only takes into account fees generated by JLP. While not practical for real-world implementation, this strategy serves as an upper bound on potential performance, helping us understand the impact of trading costs on our hedging strategies.

Each of these strategies offers a different balance between risk management and cost efficiency. Through extensive backtesting and simulation, we evaluate these strategies across various market conditions to determine the most robust and effective approach for the hJLP protocol.

6.7 Performance Metrics

We evaluate strategy performance using a comprehensive set of metrics:

- Return Metrics:
  - Compound Annual Growth Rate (CAGR)
  - Total Return
  - Alpha (excess return over unhedged JLP)
- Risk Metrics:
  - Volatility
  - Sharpe Ratio
  - Maximum Drawdown
  - Value at Risk (VaR)
  - Conditional Value at Risk (CVaR)
- Strategy-specific Metrics:

- Hedge Effectiveness (correlation between hedged and unhedged returns)
- Average Daily Delta Exposure
- Transaction Costs as a Percentage of AUM

7 Strategy Assumptions: Liquidity and Delta

In developing the hJLP strategy, we recognize that liquidity conditions in decentralized markets can be highly variable and sometimes unpredictable. To account for this, we've implemented sophisticated liquidity haircut mechanisms that allow our strategy to adapt to changing market conditions and maintain robust performance across various scenarios.

7.1 Flat Orderbook Haircut

The flat_orderbook_haircut is a straightforward yet effective method to account for potential discrepancies between simulated and real-world liquidity conditions. This haircut applies a uniform percentage decrease to the amount of liquidity available at every depth level of the order book.

Mathematically, we can express this as:

$$L_{adjusted} = L_{original} \times (1 - h_{flat}) \tag{7}$$

Where:

- $L_{adjusted}$ is the adjusted liquidity
- $L_{original}$ is the original liquidity
- h_{flat} is the flat haircut percentage

This conservative approach ensures that our strategy doesn't overestimate available liquidity, which is crucial for maintaining robustness in less liquid market conditions.

7.2 Liquidity Crunch Haircut

The liquidity_crunch_haircut is a more sophisticated approach designed to dynamically adjust liquidity estimates based on recent market conditions. This method is particularly effective in modeling liquidity during periods of market stress or unusual volatility. The process works as follows:

1. For each depth level in the order book, we compute an expanding window mean and variance of notional liquidity.
2. At each timestamp, we calculate z-scores of current liquidity with respect to these expanding window means and variances.

3. If the z-score is larger than 0.5 (indicating above-average liquidity), no haircut is applied.
4. For z-scores below 0.5, we map that z-score onto a shifted Gaussian distribution:
   - The mean of this distribution is X% smaller than the estimated mean.
   - The cumulative distribution function (CDF) at a z-score of 0.5 matches the estimated distribution.
5. Finally, we take the maximum of zero and the resulting value to ensure non-negative liquidity.

This approach allows for a more nuanced representation of liquidity, particularly during market stress events. The shifting of the Gaussian distribution for below-average liquidity conditions allows us to be more conservative in our estimates when market liquidity appears to be tightening.

7.3 Rolling Delta Hedge

This strategy uses a time-weighted average of recent deltas as the target for hedging, smoothing out short-term fluctuations and potentially reducing unnecessary trades. Implementation:

- Calculate a rolling average of delta exposures (e.g., over the past 1 hour).
- Set a threshold for deviation from this rolling average (e.g., 5% of notional value).
- When current delta deviates from the rolling average by more than the threshold, execute trades to bring delta back to the rolling average.

This strategy aims to capture longer-term trends in delta exposure while ignoring short-term noise.

8 Simulation Results

8.1 Backtesting Assumptions

The backtesting results were generated based on the following key assumptions:

These assumptions form the baseline for our backtesting scenarios, providing a consistent framework for evaluating the performance of different hedging strategies across various market conditions.

Table 1: General Backtesting Assumptions

Parameter	Value
Flat orderbook haircut	30%
Liquidity crunch haircut	20%
Perps base trading fee	5 bps
Initial JLP	$10 million notional

8.2 Backtesting Results

Based on the backtesting results, we can draw several key insights.

1. **Risk-Adjusted Returns:** All hedging strategies significantly improve risk-adjusted returns compared to the unhedged JLP, as evidenced by higher Sharpe ratios (Table 2). The "Fee only" strategy shows the highest Sharpe ratio at 28.23, but this is theoretical and doesn't account for market movements.

2. **Volatility Reduction:** Hedged strategies dramatically reduce volatility, with most strategies achieving volatility below 7% compared to the unhedged JLP's 32.17% (Table 2).

Table 2: Strategy Return and Volatility

Strategy	CAGR	Vol	Sharpe
Fee only	48.16%	1.41%	28.23
1% dollar bounds	48.65%	2.24%	17.91
5-minute hedge	38.16%	2.09%	15.67
5-minute hedge 10bps	52.52%	2.98%	14.36
20%-10% bounds	46.39%	3.57%	10.82
3%/4%/8% dollar bounds	55.65%	5.21%	8.61
Daily hedge	39.21%	6.87%	4.90
Unhedged JLP	33.97%	32.17%	1.08

Table 3: Strategy Risk Metrics

Strategy	Excess Ret	Tracking Error	Info Ratio
Fee only	0.00%	0.00%	0.00
1% dollar bounds	0.35%	2.20%	0.16
5-minute hedge	-7.06%	2.11%	-3.34
5-minute hedge 10bps	2.97%	2.25%	1.32
20%-10% bounds	-1.16%	3.74%	-0.31
3%/4%/8% dollar bounds	5.11%	5.15%	0.99
Daily hedge	-6.08%	6.68%	-0.91
Unhedged JLP	-5.05%	32.33%	-0.16

3. **Downside Protection:** Hedged strategies provide superior downside protection, with significantly improved worst 24-hour returns and maximum daily drawdowns compared to the unhedged JLP (Table 8).

4. **Trade-off between Return and Risk:** More aggressive hedging strategies (e.g., 5-minute hedge) show lower volatility but also lower returns, while less frequent hedging (e.g., daily hedge) shows higher volatility but improved PnL from perps trading (Tables 2).

5. **Transaction Costs:** The impact of transaction costs is evident in the return breakdown, with more frequent hedging strategies incurring higher perps transaction costs (Table 9).

6. **Funding Fees:** All hedging strategies benefit from positive funding fees, contributing to overall returns.

7. **Asset-Specific Bounds:** The 3%/4%/8% dollar bounds strategy shows the highest total return at 12.13%, with a good balance of perps PnL and relatively low transaction costs (Tables 9).

Overall, the 1% dollar bounds strategy emerges as strong candidates for implementation, offering a compelling balance of strong returns, reasonable volatility, and efficient use of transaction costs. The 5-minute hedge 10bps strategy also shows promise with the highest CAGR, but at the cost of slightly higher volatility and transaction costs. In general, all hedging strategies significantly reduce the delta exposure compared to the unhedged JLP. All hedging strategies significantly reduce the delta exposure compared to the unhedged JLP. The hedged deltas mostly fluctuate around zero, indicating effective risk management, as shown on Figures 3-5.

8.3 Scalability Analysis

We conducted backtests under two scenarios: lower AUM with lower fees, and higher AUM with higher fees. Key assumptions for each scenario are presented in Tables 6 and 7.

Table 4: Low AUM Scenario Assumptions

Parameter	**Value**
Initial JLP	$5 million notional
Perps base trading fee	2.5 bps
Orderbook haircut	10%

8.4 Analysis of Results

1. **5-minute hedge 10bps 1H delta**:

Table 5: High AUM Scenario Assumptions

Parameter	**Value**
Initial JLP	$100 million notional
Perps base trading fee	5 bps
Orderbook haircut	50%

- Low AUM: CAGR 49.91%, Sharpe 31.42
- High AUM: CAGR 49.39%, Sharpe 29.35
- Scalability: Good, with minimal decrease in CAGR and slight decrease in Sharpe ratio.

2. **5-minute hedge 1H delta**:
   - Low AUM: CAGR 49.91%, Sharpe 31.42
   - High AUM: CAGR 41.77%, Sharpe 28.83
   - Scalability: Moderate, with noticeable decrease in CAGR and slight decrease in Sharpe ratio.
3. **1% dollar bounds 1H delta**:
   - Low AUM: CAGR 56.76%, Sharpe 29.77
   - High AUM: CAGR 53.68%, Sharpe 28.34
   - Scalability: Good, with small decrease in both CAGR and Sharpe ratio.
4. **20%-10% bounds 1h delta**:
   - Low AUM: CAGR 61.72%, Sharpe 23.22
   - High AUM: CAGR 49.87%, Sharpe 19.68
   - Scalability: Moderate, with significant decrease in CAGR and noticeable decrease in Sharpe ratio.
5. **3%/4%/8% dollar bounds 1h delta**:
   - Low AUM: CAGR 64.79%, Sharpe 11.07
   - High AUM: CAGR 64.57%, Sharpe 11.05
   - Scalability: Excellent, with negligible decrease in both CAGR and Sharpe ratio.

8.5 Conclusion

Based on our analysis, we identify two standout strategies:

1. **Best overall balance**: 1% dollar bounds 1H delta
   - Offers good scalability, high Sharpe ratio, strong CAGR, and moderate Perps costs.
2. **Best risk-adjusted returns with scalability**: 5-minute hedge 10bps 1h delta
   - Provides high Sharpe ratio, good CAGR, and good scalability, but incurs higher Perps costs.

The 1% dollar bounds 1H delta strategy emerges as the most balanced approach, offering an optimal combination of risk-adjusted returns, costs, and absolute returns. However, the final selection may depend on specific risk tolerance, return objectives, and operational considerations of the implementation.

9 Analysis of Hedging Strategies and Backtesting Results

9.1 Hedging Strategies

We evaluated several hedging strategies for the hJLP (Hedged Jupiter Liquidity Provider) system:

1. **Bounded Delta Strategy**: Rebalance when the absolute delta exceeds a specified threshold (e.g., 10%).
2. **Rolling Delta Hedge**: Use a time-weighted average of recent deltas (e.g., 1-hour rolling average) as the hedging target, smoothing out short-term fluctuations.
3. **Asset-Specific Bounds**: Apply different delta thresholds for each asset (e.g., 3% for BTC, 4% for ETH, 8% for SOL), rebalancing when any asset exceeds its specific bound.

9.2 Backtesting Results

We conducted backtests under two scenarios: lower AUM with lower fees, and higher AUM with higher fees. Key assumptions for each scenario are presented in Tables 6 and 7.

Table 6: Low AUM Scenario Assumptions

Parameter	**Value**
Initial JLP	$5 million notional
Perps base trading fee	2.5 bps
Orderbook haircut	10%

Table 7: High AUM Scenario Assumptions

Parameter	**Value**
Initial JLP	$100 million notional
Perps base trading fee	5 bps
Orderbook haircut	50%

9.3 Analysis of Results

1. **5-minute hedge 10bps 1H delta**:
   - Low AUM: CAGR 49.91%, Sharpe 31.42
   - High AUM: CAGR 49.39%, Sharpe 29.35
   - Scalability: Good, with minimal decrease in CAGR and slight decrease in Sharpe ratio.
2. **5-minute hedge 1H delta**:
   - Low AUM: CAGR 49.91%, Sharpe 31.42
   - High AUM: CAGR 41.77%, Sharpe 28.83
   - Scalability: Moderate, with noticeable decrease in CAGR and slight decrease in Sharpe ratio.
3. **1% dollar bounds 1H delta**:
   - Low AUM: CAGR 56.76%, Sharpe 29.77
   - High AUM: CAGR 53.68%, Sharpe 28.34
   - Scalability: Good, with small decrease in both CAGR and Sharpe ratio.
4. **20%-10% bounds 1h delta**:
   - Low AUM: CAGR 61.72%, Sharpe 23.22
   - High AUM: CAGR 49.87%, Sharpe 19.68
   - Scalability: Moderate, with significant decrease in CAGR and noticeable decrease in Sharpe ratio.
5. **3%/4%/8% dollar bounds 1h delta**:

- Low AUM: CAGR 64.79%, Sharpe 11.07
- High AUM: CAGR 64.57%, Sharpe 11.05
- Scalability: Excellent, with negligible decrease in both CAGR and Sharpe ratio.

9.4 Conclusion

Based on our analysis, we identify two standout strategies:

1. **Best overall balance**: 1% dollar bounds 1H delta
   - Offers good scalability, high Sharpe ratio, strong CAGR, and moderate Perps costs.
2. **Best risk-adjusted returns with scalability**: 5-minute hedge 10bps 1h delta
   - Provides high Sharpe ratio, good CAGR, and good scalability, but incurs higher Perps costs.

The 1% dollar bounds 1H delta strategy emerges as the most balanced approach, offering an optimal combination of risk-adjusted returns, costs, and absolute returns. However, the final selection may depend on specific risk tolerance, return objectives, and operational considerations of the implementation.

9.5 Volatility Case Study and Stress Tests

The June 7 case study, as illustrated in Figure 1, demonstrates the critical importance of robust hedging strategies in volatile cryptocurrency markets. Following the release of U.S. job growth statistics at approximately 18:00 UTC, the market experienced significant turbulence, resulting in the liquidation of $84 million notional of long Jupiter perpetual futures contracts. This event caused a sudden and substantial increase in delta exposure for JLP holders, with SOL exposure doubling from 2.3 to 4.6 SOL per 1000 JLP within just 30 minutes.

To evaluate the efficacy of various hedging approaches, we conducted stress tests on five distinct strategies:

- Unhedged JLP
- 5-minute hedge
- Daily hedge at 00:00 UTC
- Daily hedge at 18:30 UTC
- 1% dollar bounds hedge

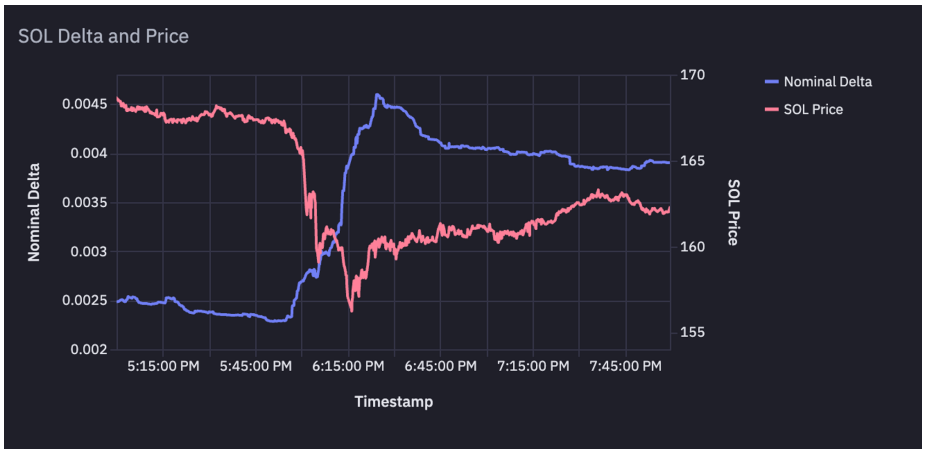

Figure 1: SOL Delta and Price

9.5.1 Analysis of Strategy Performance

1. **Unhedged JLP**: Exhibited the highest volatility, with a sharp decline in portfolio value around 06:00 PM, followed by a gradual recovery.

2. **5-minute hedge**: Demonstrated the most stable performance, maintaining a consistent portfolio value throughout the volatile period.

3. **Daily hedge (00:00 UTC)**: Showed delayed reaction to market changes, with a significant increase in portfolio value after 07:00 PM.

4. **Daily hedge (18:30 UTC)**: Performed similarly to the 1% dollar bounds strategy, with a slight decline followed by stability.

5. **1% dollar bounds**: Exhibited good stability, with only a minor decline during the initial market shock.

These stress tests revealed that while all hedging strategies mitigated the initial price drawdown to some extent, their performances diverged significantly in the aftermath of the event. Notably, the daily hedge at 00:00 UTC strategy retained considerable exposure (1.5 SOL per 1000 JLP) for an extended period, highlighting the potential risks of infrequent rebalancing in rapidly changing market conditions.

9.5.2 Market Liquidity Analysis

Our analysis of market liquidity during this volatile period showed a marked increase in slippage, with the cost of selling 20,000 SOL doubling from 50 to 100 basis points. This underscores the importance of considering liquidity constraints when implementing hedging strategies, particularly during periods of heightened market stress.

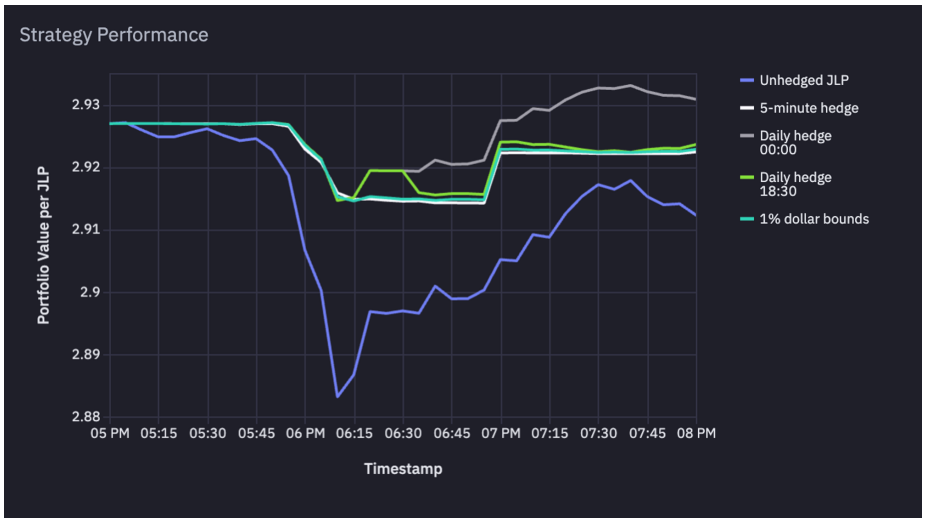

Figure 2: Strategies Performance

9.5.3 Conclusion

This case study emphasizes the need for dynamic and responsive hedging mechanisms in cryptocurrency markets. It also highlights the complex interplay between hedging frequency, market liquidity, and overall portfolio performance in the face of sudden market shocks. The 5-minute hedge and 1% dollar bounds strategies demonstrated superior performance in maintaining portfolio stability during extreme market conditions, suggesting their potential effectiveness for the hJLP system.

10 Supporting Data

Table 8: Strategy Drawdown Metrics

Strategy	**Worst 24h**	**Max Daily DD**
Fee only	0.62%	0.00%
1% dollar bounds	0.53%	-0.30%
5-minute hedge	0.20%	-0.36%
5-minute hedge 10bps	0.39%	-0.11%
20%-10% bounds	-0.48%	-1.46%
3%/4%/8% dollar bounds	-0.73%	-1.89%
Daily hedge	-2.86%	-2.97%
Unhedged JLP	-14.23%	-9.79%

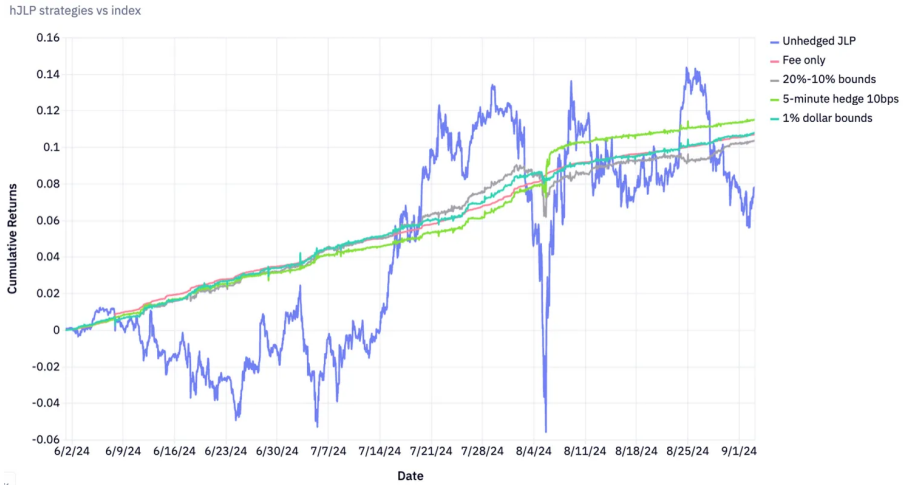

Figure 3: hJLP Strategies vs Index

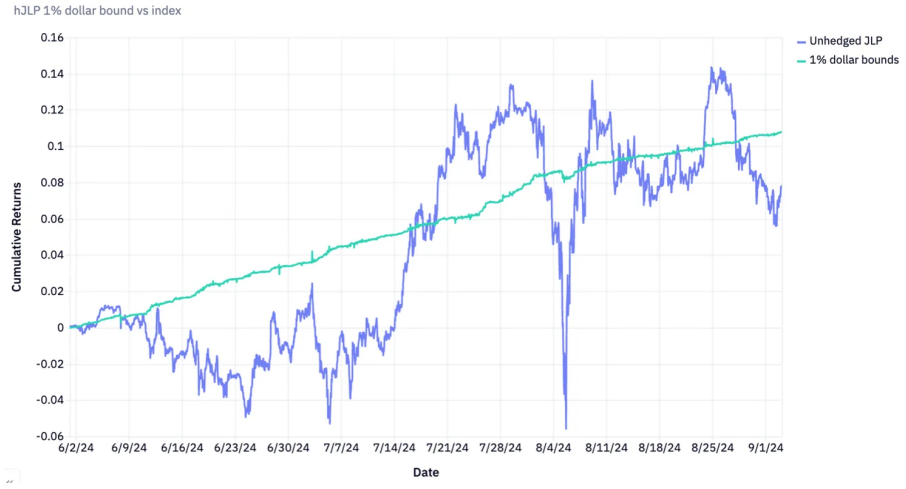

Figure 4: hJLP 1 pct.dollar bound vs Index

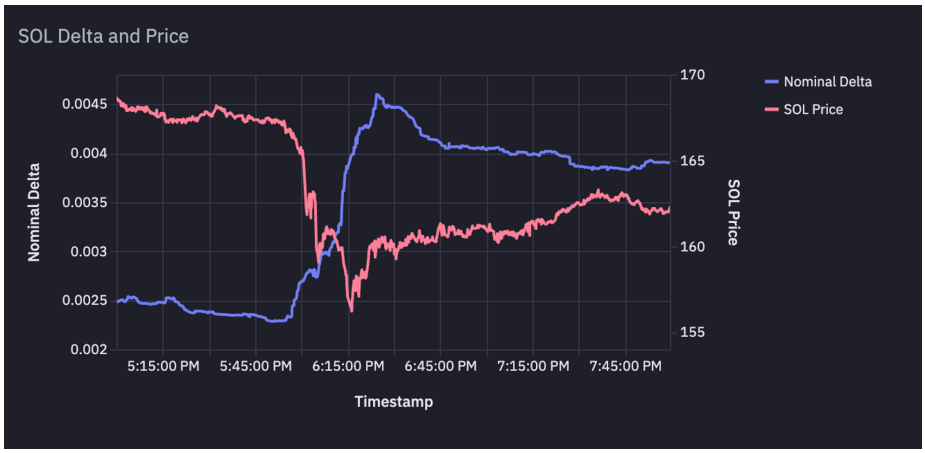

Figure 5: Delta Paths - BTC

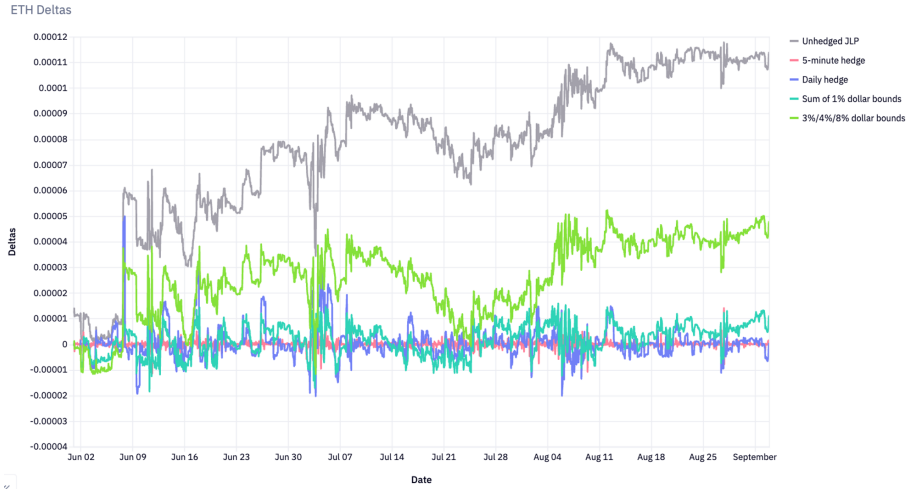

Figure 6: Delta Paths - ETH

Figure 7: Delta Paths - SOL

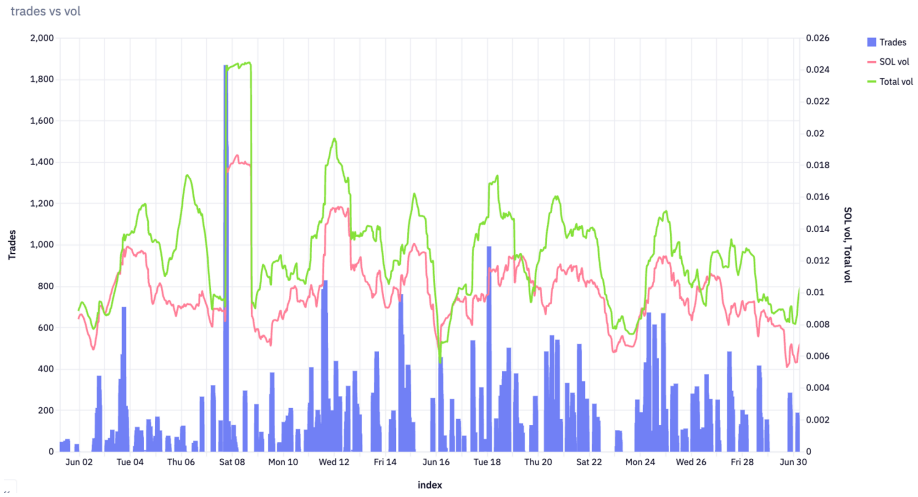

Figure 8: Trades and Volatility

Table 9: Strategy Total Return and Costs

Strategy	**Total Return**	**Perps tcosts**	**JLP spot tcosts**
Unhedged JLP	7.91%	0.00%	0.00%
5-minute hedge	8.73%	-1.86%	-0.62%
5-minute hedge 10b	11.54%	-1.24%	-0.62%
Daily hedge	8.96%	-5.57%	-0.04%
20%-10% bounds	10.37%	-1.49%	-0.61%
1% dollar bounds	10.81%	-0.63%	-0.62%
3%/4%/8% dollar b	12.13%	-0.14%	-0.58%

11 Further Reading and Resources

For those interested in delving deeper into the theoretical foundations and practical implementations of portfolio management strategies similar to hJLP, we recommend the following resources:

- Optimal Portfolio Management with Transactions Costs and Capital Gains Taxes: This paper provides insights into optimizing portfolio management while considering real-world constraints such as transaction costs and taxes.
- Multiperiod Portfolio Optimization with General Transaction Costs: An in-depth exploration of portfolio optimization strategies across multiple time periods, taking into account various types of transaction costs.

For practical implementation and integration with the Drift protocol:

- Drift Trading Platform: The main interface for interacting with Drift's perpetual futures markets.
- Drift Documentation: Comprehensive documentation covering all aspects of the Drift protocol.
- Drift SDK Documentation: Detailed guide for developers looking to integrate Drift's functionality into their applications.
- Drift Labs GitHub: Open-source repositories containing Drift's smart contracts and related tools.